

Rimjhim Saxena

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ACADEMIC APPOINTMENTS

Postdoctoral Researcher, University of Geneva

Sept 2025 - Present

Institute of Economics and Econometrics
Geneva School of Economics and Management

EDUCATION

Ph.D. Economics, University of Colorado Boulder

2025

Disseration: "Three Essays in the Economics of Climate Change"

Advisor: Jonathan Edward Hughes, Committee: Daniel Kaffine, Taylor Jaworski, Beia Spiller

M.A. Economics, University of Colorado Boulder

2021

M.A. Economics, Boston University

2017

M.B.A. Finance, Thapar Institute of Engineering & Technology

2014

B.E. Industrial Engineering, Thapar Institute of Engineering & Technology

2014

RESEARCH AND TEACHING INTERESTS

Primary: Environmental Economics, Development Economics, Economics of Climate Change

Secondary: International Trade, Applied Microeconomics, Political Economics

WORKING PAPERS

1. *Barriers within Borders: Climate Change, Market Barriers, and the Reversal of Structural Transformation*
([Job Market Paper](#))

This paper examines how climate change is hindering structural transformation in India and the role of internal trade barriers in exacerbating the effect. I combine local temperature effects on productivity, consumption, and labor shares to evaluate the potential for labor reallocation out of agriculture as an adaptation to climate change. Empirical findings show that temperature has a more negative impact on agricultural productivity versus manufacturing (supply-side effect), and household expenditure on food increases as incomes fall, consistent with Engel's law (demand-side effect). Overall, in equilibrium, rising temperatures increase the agricultural labor share. Therefore hindering structural transformation. Theoretically, districts in India could mitigate against the adverse effects of declining agricultural productivity through inter-district trade with less affected districts. Using a market access variable created by using development of Indian highways, I find that improved market access does not disrupt the positive relationship between temperature and agricultural labor share due to existing barriers in trade within India. These trade barriers reduce spatial competition among agricultural buyers and traps labor in low-productivity agriculture. I then use a spatial equilibrium model with internal trade barriers for counterfactual analysis which reveals that removing state-level trade barriers in Indian agriculture would decrease agricultural labor share by 11.05% on average for each district (~ 28.9 million people across India).

2. *Stumped by the sun, Saved by the side: How do teams adapt to heat? Evidence on Peer Adaptation*

How do teams respond to temperature when individual productivity declines? A large body of literature documents that heat deteriorates individual performance. However, workers rarely work in isolation. Using high-frequency data from international cricket games, this paper examines how heat affects team production. I first document a decline in individual productivity of about 1.46% - 2.68% for every degree Celsius increase in temperature, consistent with the literature. In contrast, I find that equilibrium output remains unaffected by heat. To understand this puzzle, I decompose equilibrium outcomes into individual, peer, and adversarial components and introduce a temperature-based split-sample variance decomposition to compare these components across hotter and cooler environments. The results show that the importance of peer interactions increases significantly under heat, while adversarial interactions remain unchanged. Further analysis shows that this adaptation operates through complementarity in skills. Back-of-the-envelope calculation finds that complementary peer interaction compensates about 45% of the decline in individual productivity.

WORKS IN PROGRESS

1. *Quantifying global trade leakage under the EUDR policy* (with [Matthieu Stigler](#))
 2. *Climate Change and Mechanization of Agriculture in India* (with [Bhavya Srivastava](#))
 3. *Climate Change and Informal Insurance Networks* (with [Matthew Gordon](#))
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PUBLICATIONS

BOOK CHAPTER

Chatterjee, D., Lal, B., & Saxena, R. (2021). [Education: Understanding the Gender Gap in Education and Employment](#). In S. Chakravorty & N. Sircar (Eds.), *Colossus: The Anatomy of Delhi* (South Asia in the Social Sciences, pp. 273-289). Cambridge: Cambridge University Press. [doi:10.1017/9781108935654.013](#)

OTHER WRITING

Arora, K., Saxena, R. (2018). [India-China Economic Relations: An Assessment](#). *South Asian Voices*, Stimson Center, Washington D.C

TEACHING EXPERIENCE

GSEM, University of Geneva

Guest Lecturer: Environmental Economics and Climate Change [Masters Level] (~30 students) Sept 2025 - Dec 2025

University of Colorado Boulder

Instructor of Record: Natural Resource Economics (~50 students) Aug 2024 - Dec 2024

Instructor of Record: Principles of Microeconomics (~200 students) Aug 2022 - Dec 2022

Managed a team of four teaching assistants

Teaching Assistant Aug 2019 - May 2025

Principles of Microeconomics (x5 semesters)

Principles of Macroeconomics (x4 semesters)

Environmental Economics (x2 semesters)

Introduction to Statistics with Computer Applications (x1 semester)

AWARDS

Heartland Environmental and Resource Economics Workshop Travel Support, 2023
TWEEDS workshop Travel Grant, 2023 (\$525)
Faculty Support Fellowship, 2023 (\$515)
Graduate School Domestic Travel Grant, 2023 (\$450)
Graduate Award for Public Policy Research, 2022 (\$2,000)
Reuben A. Zubrow Fellowship in Economics, 2021 (\$2,000)
Continuing Education Teaching Fellowship, University of Colorado Boulder, 2021 (\$2,000)
Continuing Education Teaching Fellowship, University of Colorado Boulder, 2020 (\$1,000)
Scholarship, Department of Economics, University of Colorado Boulder, 2019 (\$10,000)

SEMINAR PRESENTATIONS

2026: Paris Environmental Economics Workshop (*April, Scheduled*); International Energy Workshop, Cape Town, South Africa (*June, Scheduled*)

2025: AERE Session at Eastern Economic Association Conference, New York NY; Seton Hall University; Truman State University; IIT Gandhinagar; AERE Summer Conference (*Scheduled*); GSEM, University of Geneva

2024: 25th Annual CU Environmental and Resource Economics Workshop, Vail CO; DARE Seminar at Colorado State University; AERE Session at Southern Economic Association Conference, Washington D.C.

2023: 18th Annual Conference on Economic Growth and Development, Indian Statistical Institute, Delhi; Heartland Environmental and Resource Economics Workshop; 2023 Berkeley/Sloan Summer School in Environmental and Energy Economics; 98th Western Economic Association Conference; AERE Summer Conference; 12th Annual Front Range Energy and Environmental Economics Camp

2022: 23rd Annual CU Environmental and Resource Economics Workshop; 11th Annual Front Range Energy and Environmental Economics Camp

2021: Environmental Defense Fund

2019: Centre for Policy Research

PROFESSIONAL SERVICE

Association of Environmental and Resource Economists (AERE)

<i>Organizing Committee</i> , AERE sessions at Southern Economic Conference	2024
<i>Session Chair</i> , AERE sessions at Southern Economic Conference	2024

University of Colorado Boulder

<i>Organizer</i> , Economics Graduate Student Seminar	2021 - 2025
<i>Host</i> , Graduate Student Panel at Graduate Student Recruitment	Spring 2024
<i>Mentor</i> , Graduate student Mentor (x2)	2021 - 2023

PROFESSIONAL MEMBERSHIPS

Association of Environmental and Resource Economists (AERE), Western Economic Association International (WEAI)

OTHER WORK EXPERIENCE

University of Colorado Boulder

Research Assistant (Dr. Taylor Jaworski)

May 2022 - July 2022

Digitized manufacturing plant opening locations in US during World War II
Created data visualizations, and spatial equilibrium model

Research Assistant (Dr. Francisca Antman)

May 2020 - July 2020

Antman, F. M., & Cortes, K. E. (2023). The Long-Run Impacts of Mexican American School Desegregation. *Journal of Economic Literature*, 61(3), 888-905.

Environmental Defense Fund

Pre Doctoral Fellow

June 2021 - Aug 2021

Under Office of the Chief Economist

Collaborated with Economics, Science, Policy teams in USA and India on research on Economic impacts of Environment, and Environmental justice questions

University of Pennsylvania

Research Associate, Centre for Advanced Study of India (CASI)

Jan 2018 - July 2019

In collaboration with Centre for Policy Research (CPR), Delhi

Vaishanv, Milan (2021). [Gender, social change, and urbanisation in North India](#). *International Growth Centre Project #IND-19069*

Khosla, R., Sircar, N., & Bhardwaj, A. (2019). Energy demand transitions and climate mitigation in low-income urban households in India. *Environmental Research Letters*, 14(9), 095008.

Bhardwaj, A., Joshi, M., Khosla, R., & Dubash, N. K. (2019). More priorities, more problems? Decision-making with multiple energy, development and climate objectives. *Energy Research & Social Science*, 49, 143-157.

[Small City Dreaming Project](#), Centre for Policy Research

SKILLS

Proficient: R, Julia, Quarto Markdown, GIS Tools, L^AT_EX, C, C#

Intermediate: Python, Stata, Matlab, SQL

Languages: English (Native), Hindi (Native), Urdu, Punjabi, Italian (B1)

REFERENCES

Jonathan Edward Hughes (advisor)

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Taylor Jaworski (committee member)

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Daniel Kaffine (committee member)

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Beia Spiller (committee member)

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